



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE:ADDAPTIVE SOFTWARE ENGINEERING

A.Y. 2026 – 2027

TITLE OF THE PROJECT	AI-Based Library Management & Recommendation System	
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Abstract

Libraries traditionally depend on manual processes to manage books, members, book issues, returns, and fines. These processes can be time-consuming and may lead to problems such as misplaced records, incorrect information, delays in updating book availability, and difficulty in tracking user transactions. To overcome these challenges, our project, AI-Based Library Management & Recommendation System, is designed to digitize and automate the complete library management process. The system provides a simple and user-friendly web interface through which librarians can easily manage books, register members, issue and return books, maintain transaction records, and calculate fines. This reduces the amount of manual work required and makes the overall library management process faster, more accurate, and efficient.

The major feature that makes our project different from a traditional library management system is the AI-based book recommendation system. The recommendation module is developed using Python and analyzes the user's previous borrowing history and interests to suggest books that they may find useful or interesting. For example, if a user frequently borrows books related to artificial intelligence or programming, the system can recommend other books from similar categories. This provides a personalized experience for every user and encourages them to explore more books. In this way, the system not only manages library operations but also helps users discover books based on their individual interests.

From the technical perspective, React.js is used to develop the frontend and provide a fast and responsive user interface. Node.js is used for the backend to handle authentication, business logic, and communication between the frontend and database. MySQL is used to store important information such as book details, member information, borrowing records, returns, and fines, while Python is used for implementing the AI recommendation component. By combining web technologies with artificial intelligence, our system provides a complete and modern solution for library management. Overall, the project helps librarians reduce manual effort and maintain accurate records while providing users with a convenient and personalized library experience.